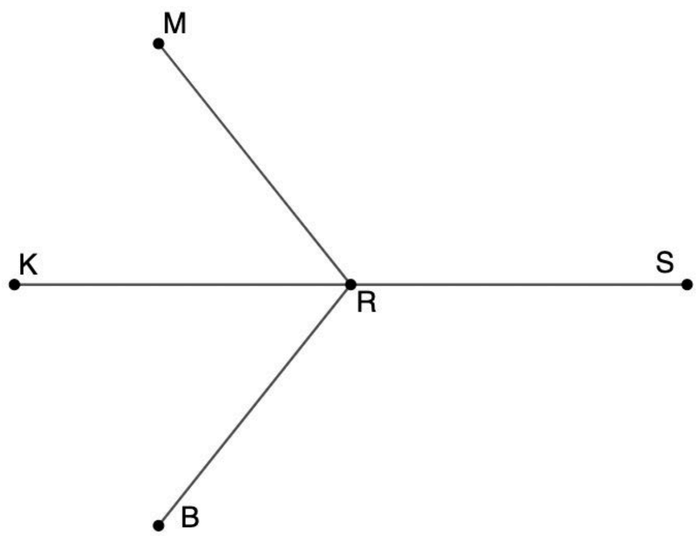
Geometry	Name____________________
Unit 3	Date____________________
Lesson 1 Homework	Period____________________

Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank	
Definition of Linear Pair	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

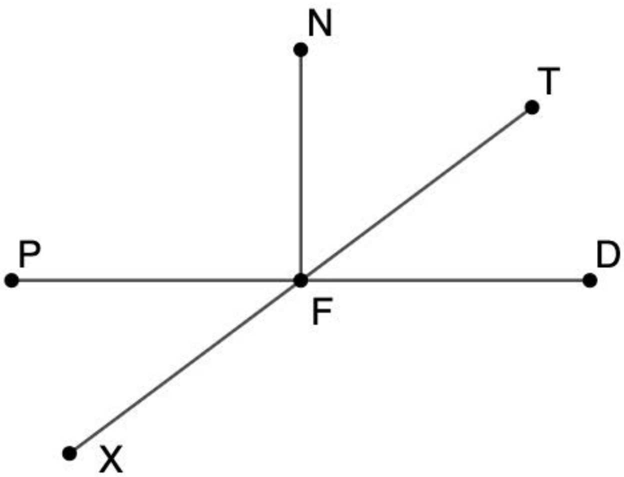
Given: $\angle KRB \cong \angle KRM$
Prove: $\angle BRS \cong \angle MRS$



Complete the table using the word bank to prove $\angle BRS \cong \angle MRS$.

Statement	**Reason**
$\angle KRB \cong \angle KRM$	Given
$\angle KRB$ and $\angle SRB$ form a linear pair	1.
$\angle KRM$ and $\angle MRS$ form a linear pair	2.
$\angle KRB$ and $\angle SRB$ are supplementary angles	3.
$\angle KRM$ and $\angle MRS$ are supplementary angles	4.
$\angle BRS \cong \angle MRS$	5.

Line segments $\overline{PD}$, $\overline{NF}$, and $\overline{TX}$ intersect at point F as shown where $m\angle PFN = 90°$, $m\angle XFP = (3x - 56)°$, $m\angle NFT = (4z + 7)°$ and $m\angle TFD = (x + 2)°$.



6. Find the value of x.

7. Find the value of z.

8. What is $m\angle NFT$?

9. What is $m\angle XFD$?

Geometry
Unit 3
Lesson 2 Homework

Name____________________
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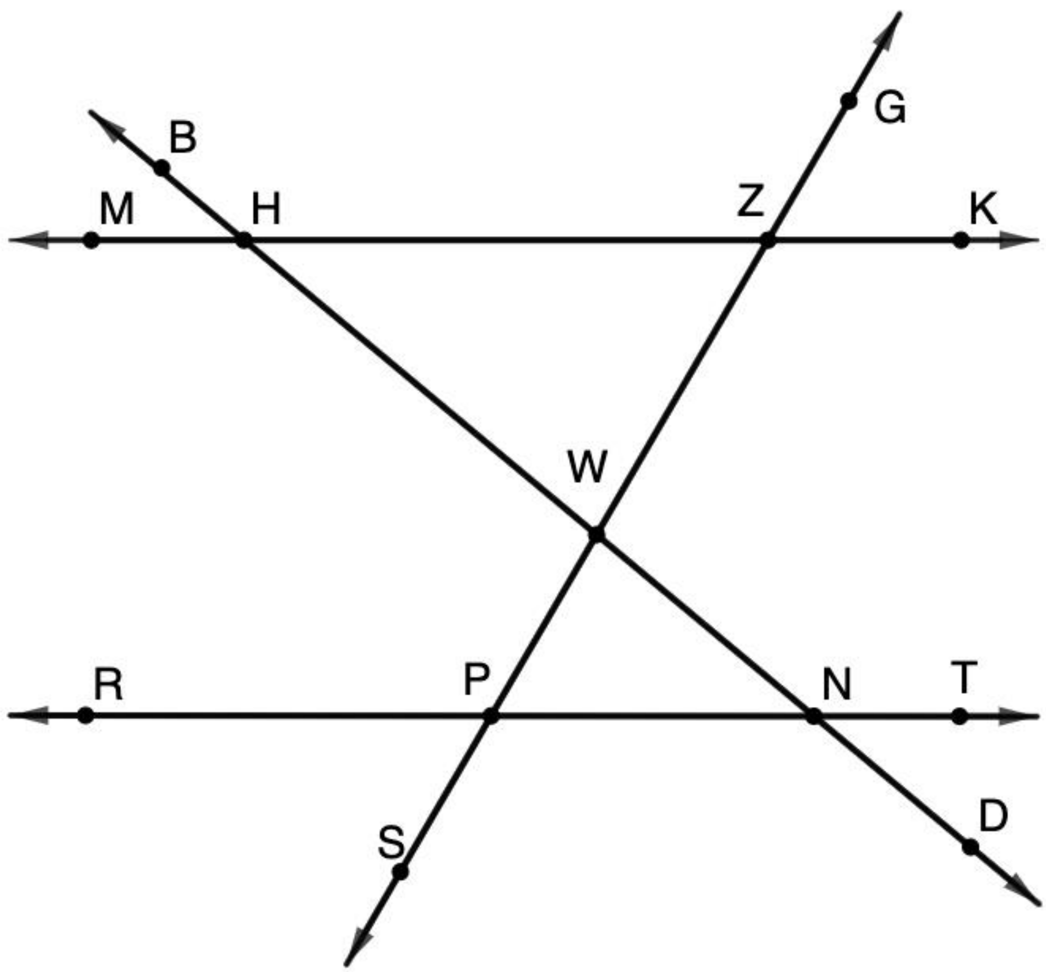
$\overleftrightarrow{MK}, \overleftrightarrow{RT}, \overleftrightarrow{BD}$ and $\overleftrightarrow{GS}$ are shown where $\overleftrightarrow{MK} \parallel \overleftrightarrow{RT}$, $\overleftrightarrow{BD} \perp \overleftrightarrow{GS}$, $m\angle MHB = 34°$, $m\angle GZK = 56°$, $m\angle WHZ = (9x - 2)°$, and $m\angle WPN = (8y)°$.



1. What is the value of x?

2. What is the value of y?

Determine the measure of the angles.

3. $m\angle WHZ$

4. $m\angle WZH$

5. $m\angle HWZ$

6. $m\angle RPS$

7. $m\angle TND$

8. $m\angle HNT$

9. $m\angle PNT$

10. $m\angle SPN$

Geometry
Unit 3
Lesson 3 Homework

Name____________________
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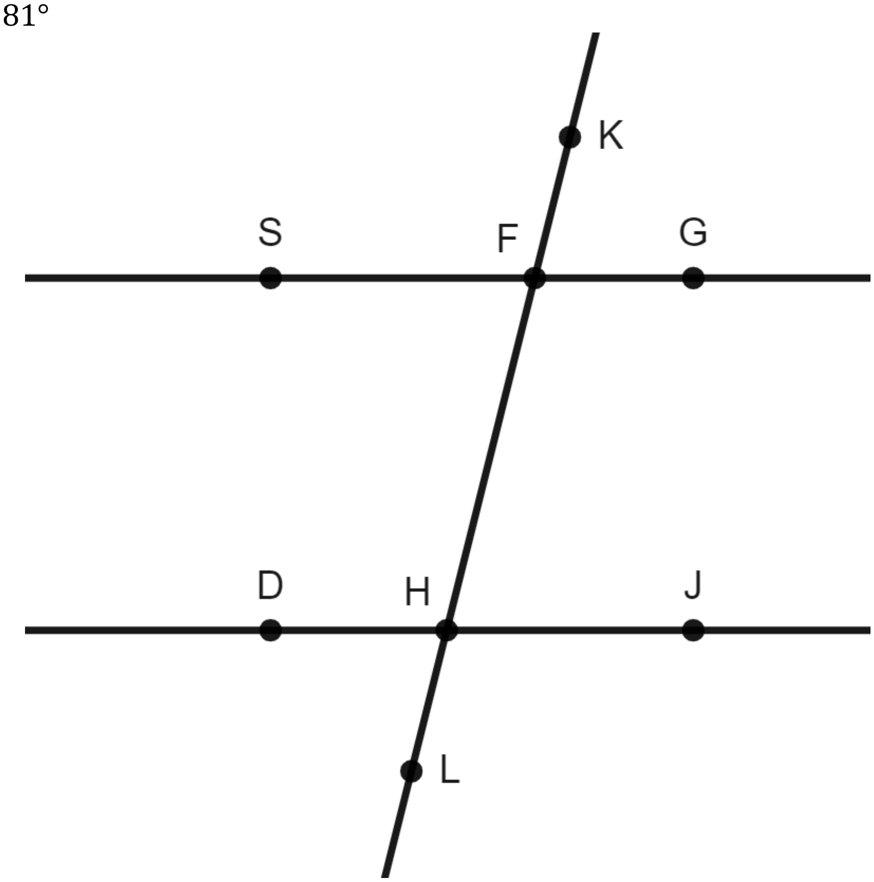
Lines $\overline{SG}$, $\overline{DJ}$, and $\overline{KL}$ are shown.
$\overline{SG} \parallel \overline{DJ}$, $m\angle FHJ = 81°$



Complete the table.

Angle	**Angle Measure**	**Relationship with $\angle FHJ$**
$\angle DHL$	1.	2.
$\angle KFG$	3.	4.
$\angle SFH$	5.	6.

$\angle FHJ$ and $\angle FHD$ are supplementary, so $m\angle FHD = 99°$.

Complete the table.

Angle	Angle Measure	Relationship with $\angle FHD$
$\angle LHJ$	7.	8.
$\angle KFS$	9.	10.

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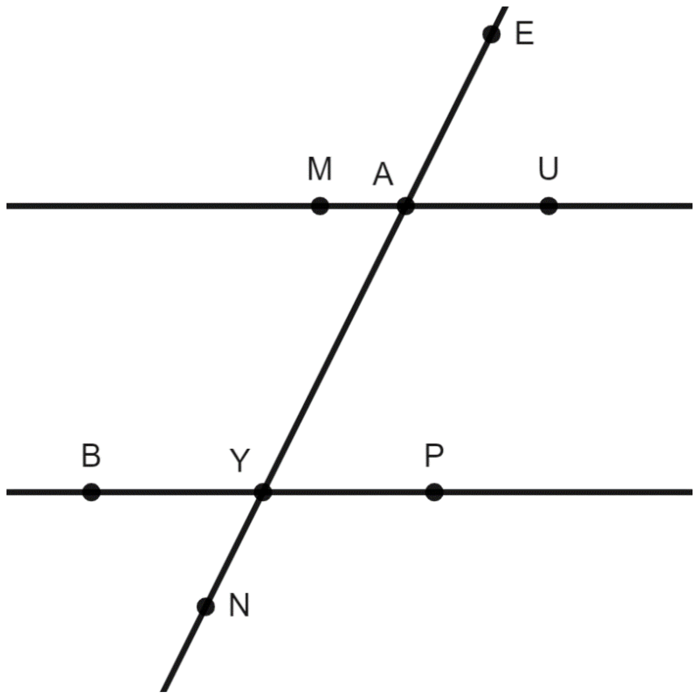
Lesson 4 Homework Period______________________

Lines $\overline{\mathrm{MU}}$, $\overline{\mathrm{BP}}$, and $\overline{\mathrm{NE}}$ are shown.
$\overline{\mathrm{MU}} \parallel \overline{\mathrm{BP}}$, $\angle\mathrm{EAU} = 63°$



Identify the relationship between each pair of angles.

Angles	Congruent or Supplementary?
$\angle\mathrm{EAU}$ and $\angle\mathrm{YAU}$	1.
$\angle\mathrm{EAU}$ and $\angle\mathrm{PYN}$	2.
$\angle\mathrm{EAU}$ and $\angle\mathrm{MAE}$	3.
$\angle\mathrm{EAU}$ and $\angle\mathrm{MAY}$	4.

Complete the table according to the diagram.

Angle	Angle Measure	Relationship with $\angle MAY$
$\angle BYA$	5.	6.
$\angle MAE$	7.	8.

9. If $\angle EAU = (7x - 14)°$ and $\angle PYN = (11x - 4)°$, what is the value of x?

10. If $\angle MAY = (4w - 5)°$ and $\angle BYA = (6w + 15)°$, what is the value of w?

Geometry Name____________________
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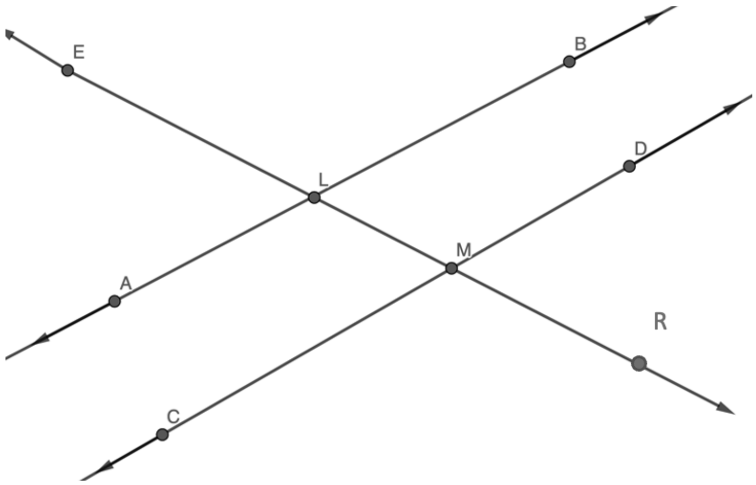
Lesson 5 Homework Period____________________

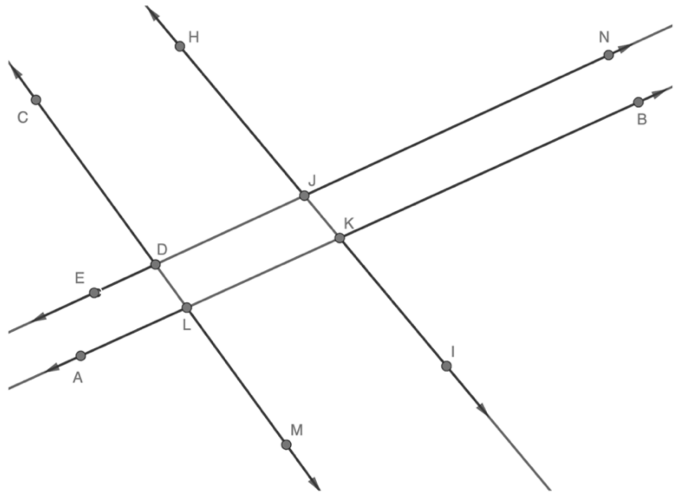
1. Given $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{ER}$. $\overleftrightarrow{ER}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$.



Determine if the given condition could be used to justify that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$?	Justification
$\angle ELA \cong \angle DMR$	o Yes o No	
$\angle ALM \cong \angle BLE$	o Yes o No	
$m\angle CMR + m\angle RMD = 180°$	o Yes o No	
$\angle ELB \cong \angle LMD$	o Yes o No	
$\angle ELB \cong \angle RMD$	o Yes o No	
If $m\angle ELB = 102°$, then $m\angle ELA = 78°$	o Yes o No	
If $m\angle RMC = 91°$, then $m\angle ALM = 89°$	o Yes o No	

2. $\overleftrightarrow{CM}$, $\overleftrightarrow{HI}$, $\overleftrightarrow{EN}$, and $\overleftrightarrow{AB}$ are shown where $\angle LKI = 94°$.



Determine if the given conditions could be used to justify that $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$. Then justify your answer.

Condition	Is $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$?	Justification
$\angle CDJ \cong \angle KLM$	o Yes o No	
$\angle CDE \cong \angle MLA$	o Yes o No	
$m\angle EDL + m\angle NJK = 180°$	o Yes o No	
$\angle KLM \cong \angle HJN$	o Yes o No	
$\angle JKL \cong \angle BKI$	o Yes o No	
If $\angle BKI = 86°$, then $m\angle BKI + m\angle LKI = 180°$	o Yes o No	

Geometry
Unit 3
Lesson 6 Homework

Name_____________________
Date______________________
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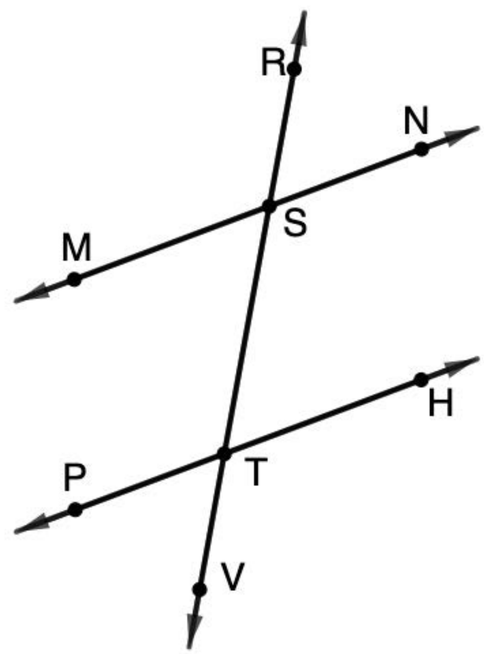
$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$, where $\overleftrightarrow{RV}$ is a transversal with $m\angle RSM = (12x - 7)°$, and $m\angle PTV = (4x - 5)°$ as shown.



1. Determine the value of x.

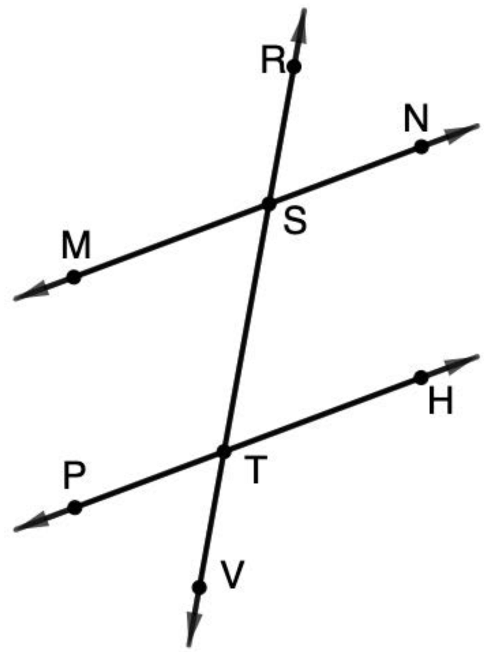
2. What is $m\angle RSM$?

3. What is $m\angle PTV$?

4. List all the angles congruent to $\angle RSM$.

5. List all the angles congruent to $\angle PTV$.

$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$, where $\overleftrightarrow{RV}$ is a transversal with
$m\angle RSN = (3x - 17)°$ and $m\angle STP = (7x - 3)°$ as shown.



6. Determine the value of x.

7. What is $m\angle RSN$?

8. What is $m\angle STP$?

9. What is $m\angle RSM$?

10. What is $m\angle HTS$?

Geometry
Unit 3
Lesson 7 Homework

Name____________________
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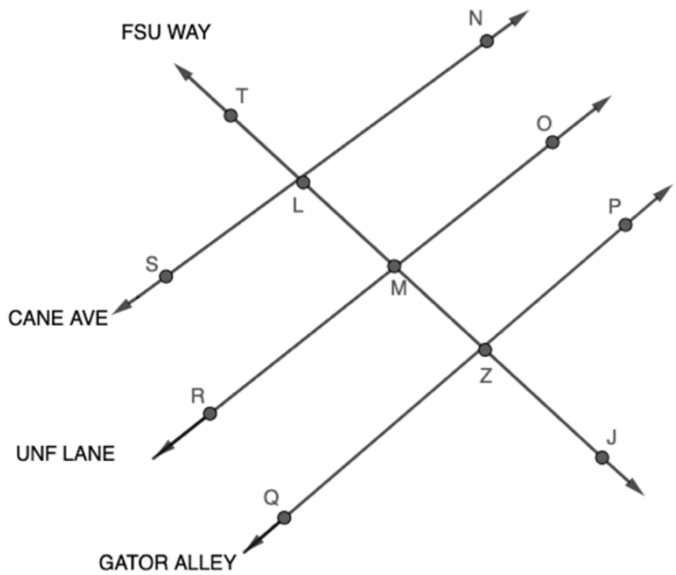
Use the following information for questions 1-3. FSU Way, a transversal, TJ, crosses the streets Cane Avenue, UNF Lane, and Gator Alley.



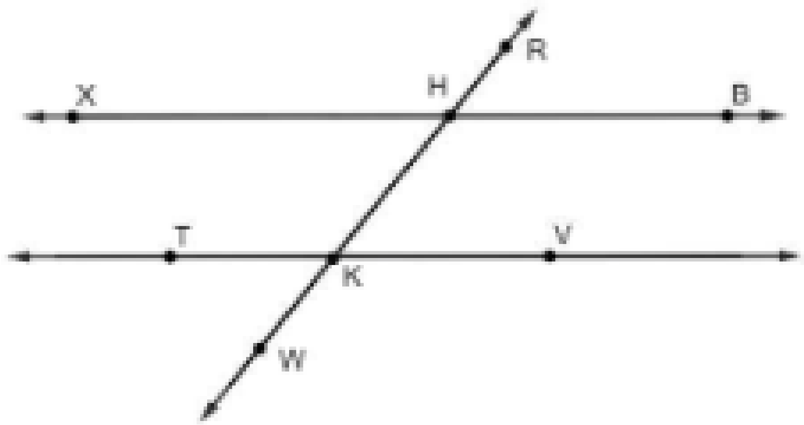
Determine the value of the variables and each angle measure given Cane Avenue, UNF Lane, and Gator Alley are parallel to each other.

1. $\angle TLN = (3x - 18)^\circ$, $\angle MZQ = (5x + 14)^\circ$, and $\angle NLM = y^\circ$

2. $\angle JZQ = (9x + 9)^\circ$ and $\angle TLN = (8x + 19)^\circ$

3. $\angle ZMR = (x + 14)^\circ$, $\angle LMO = (2x - 6)^\circ$, and $\angle LMR = y^\circ$

Use the following information for questions 4-7. Given $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal.



Determine the value of the variables and each angle measure.

4. $m\angle TKW = (8x - 20)°$ and $m\angle XHK = (3x + 2)°$

5. $m\angle RHB = (2x - 20)°$ and $m\angle XHK = 44°$

6. $m\angle HKT = (3y - 18)°$, $m\angle VKW = (8x + 12)°$, and $m\angle RHB = (2x + 18)°$

7. $m\angle BHK = (8x + 12)°$ and $m\angle VKW = 140°$

Geometry
Unit 3
Lesson 8 Homework

Name_____________________
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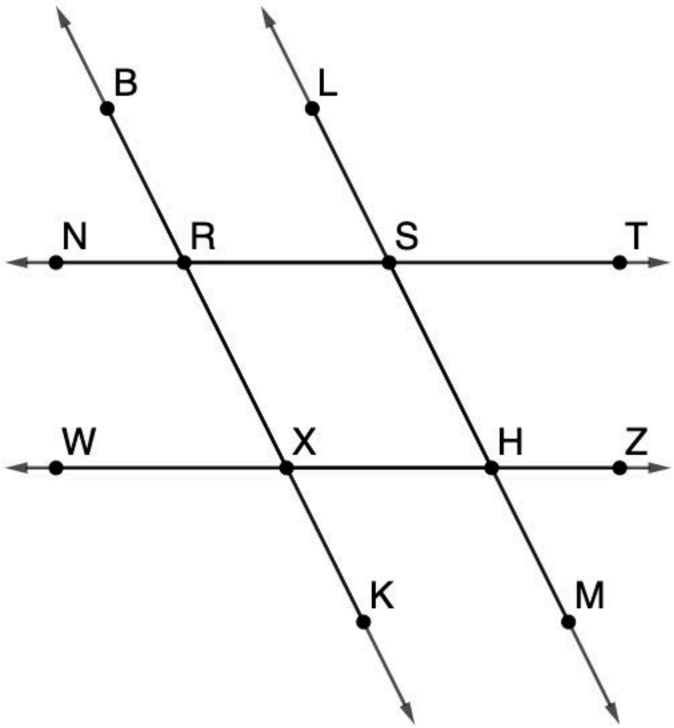
Lines $\overleftrightarrow{BK}$, $\overleftrightarrow{LM}$, $\overleftrightarrow{NT}$ and $\overleftrightarrow{WZ}$ are shown.



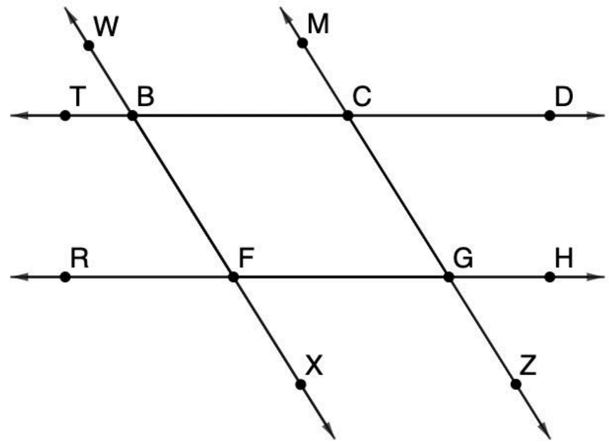
Circle the contradictory piece of information in each row.

	Item 1	**Item 2**	**Item 3**	**Item 4**
1.	$\angle RSL \cong \angle XHM$	$\angle BRN \cong \angle XRS$	$\angle WXK$ and $\angle NRB$ are supplementary	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$
2.	$\overleftrightarrow{BK} \parallel \overleftrightarrow{LM}$	$\angle LST$ and $\angle NRX$ are supplementary	$m\angle LST = 120°$	$\angle WXK \cong \angle SHZ$
3.	$\angle LST$ and $\angle WXR$ are supplementary	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$	$\overleftrightarrow{BK} \parallel \overleftrightarrow{LM}$	$m\angle BRS = 195°$
4.	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$	$\angle RSL \cong \angle HST$	$\overleftrightarrow{BK} \nparallel \overleftrightarrow{LM}$	$\angle NRB \cong \angle MHZ$
5.	$\angle NRX \cong \angle LST$	$\angle NRB$ and $\angle MHZ$ are supplementary	$\angle LST$ and $\angle MHZ$ are supplementary	$\angle RSH \cong \angle HXR$

Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\angle MCD \cong \angle CGH$

Prove: $\angle CBF \cong \angle XFG$



It is given that (6)__________________. Since the (7)________________________________, it must be true that (8)__________________. If two parallel lines are intersected by a transversal, consecutive angles are supplementary, then it must be true that (9) ___________________ are supplementary. Angles $\angle BFG$ and $\angle XFG$ are supplementary because they form (10) __________________. Since supplements of the same angle are congruent, it must be true that $\angle CBF \cong \angle XFG$.

Statements and Reasons	
$\angle MCD \cong \angle CGH$	vertical angles
$\angle MCD \cong \angle RFX$	a linear pair
$\angle FGZ \cong \angle ZGH$	consecutive angles are supplementary
$\overleftrightarrow{TD} \| \overleftrightarrow{RH}$	corresponding angles are congruent
$\overleftrightarrow{WX} \| \overleftrightarrow{MZ}$	alternate interior angles are congruent
$\angle TBW$ and $\angle MCD$	converse of the corresponding angle theorem
$\angle CBF$ and $\angle BFG$	
$\angle WBC$ and $\angle XFG$	

Geometry
Unit 3
Lesson 9 Homework

Name______________________
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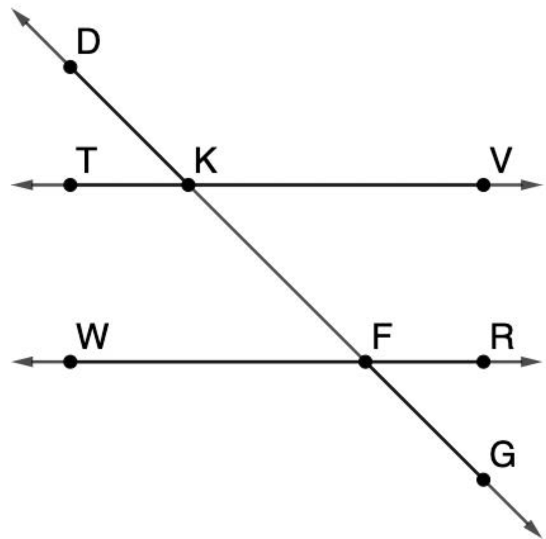
Complete the two-column proof.
Given: $\angle TKF \cong \angle KFR$

Prove: $\angle VKD \cong \angle WFG$



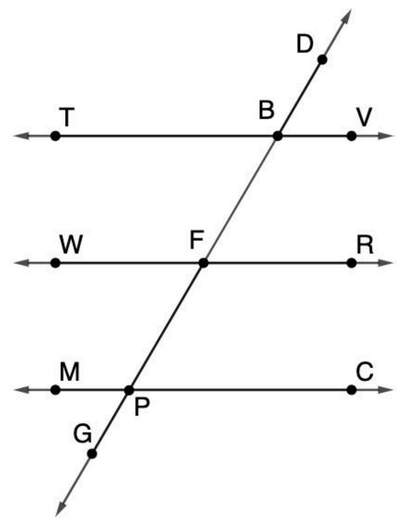
Statement	Reason
$\angle TKF \cong \angle KFR$	Given
1.	2.
3.	4.

Write the statements and reasons in the correct order in the two-column proof.

$\angle TKD \cong \angle GFR$	Supplements of congruent angles are congruent.
$\angle VKD \cong \angle WFG$	If two angles form a linear pair, then they are supplementary
$\overleftrightarrow{TV} \| \overleftrightarrow{WR}$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle GFR$ and $\angle WFG$ are supplementary	If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.
Given: $\angle DBV \cong \angle MPG$ and $\angle TBD \cong \angle WFB$

Prove: $\angle RFP \cong \angle MPF$



Statement	Reason
$\angle DBV \cong \angle MPG$	Given
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

Write the statements and reasons in the correct order in the two-column proof.

$\angle PFR \cong \angle MPF$	Given
$\angle TBD \cong \angle WFB$	Transitive Property of Congruence
$\angle PFR \cong \angle TBD$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle TBD \cong \angle MPF$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\overleftrightarrow{TV} \| \overleftrightarrow{MC}$	If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
$\overleftrightarrow{TV} \| \overleftrightarrow{WR}$	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.